

Activation of mutant *TERT* promoter by RAS-ERK signaling is a key step in malignant progression of BRAF-mutant human melanomas

Yinghui Li^{a,1}, Hui Shan Cheng^a, Wee Joo Chng^{b,c,d}, and Vinay Tergaonkar^{a,e,f,g,1}

^aInstitute of Molecular and Cell Biology, Agency for Science, Technology and Research, Singapore 138673; ^bCancer Science Institute of Singapore, National University of Singapore, Singapore 117599; ^cDepartment of Medicine, Yong Loo Lin School of Medicine, National University of Singapore, Singapore 117597; ^dDepartment of Haematology–Oncology, National University Cancer Institute of Singapore, National University Health System, Singapore 117597; ^eDepartment of Biochemistry, Yong Loo Lin School of Medicine, National University of Singapore, Singapore 117597; ^fCentre for Cancer Biology, University of South Australia, Adelaide, SA 5000, Australia; and ^gInflammation and Human Ailments Laboratory, SA Pathology, Adelaide, SA 5000, Australia

Edited by James E. Cleaver, University of California, San Francisco, CA, and approved October 31, 2016 (received for review July 7, 2016)

Although activating BRAF/NRAS mutations are frequently seen in melanomas, they are not sufficient to drive malignant transformation and require additional events. Frequent co-occurrence of mutations in the promoter for *telomerase reverse transcriptase* (*TERT*), along with BRAF alterations, has recently been noted and correlated with poorer prognosis, implicating a functional link between BRAF signaling and telomerase reactivation in melanomas. Here, we report that RAS-ERK signaling in BRAF mutant melanomas is critical for regulating active chromatin state and recruitment of RNA polymerase II at mutant *TERT* promoters. Our study provides evidence that the mutant *TERT* promoter is a key substrate downstream of the RAS-ERK pathway. Reactivating *TERT* and hence reconstituting telomerase is an important step in melanoma progression from nonmalignant nevi with BRAF mutations. Hence, combined targeting of RAS-ERK and *TERT* promoter remodeling is a promising avenue to limit long-term survival of a majority of melanomas that harbor these two mutations.

TERT promoter mutations | BRAF mutations | ERK-MAPK pathway | telomerase reactivation | cancer

Human melanoma is a major skin malignancy accounting for nearly 80% of deaths linked to skin cancers worldwide (1). Due to the high metastatic potential of melanomas and the increased resistance of malignant melanomas to conventional therapies, melanoma is well known for its aggressiveness and poor survival rates (2, 3).

The RAS-RAF-MEK-ERK signaling pathway is one of the key oncogenic pathways that drives melanoma progression and is also known to play a central role in normal melanocyte differentiation and expansion (4). Pathological mutations in the genes encoding protein kinases that participate in the RAS-RAF-MEK-ERK pathway were identified more than a decade ago (5–7). In particular, oncogenic mutations in the B-Raf proto-oncogene, serine/threonine kinase (BRAF), which signals upstream of MEK, as well as the GTPase protein NRAS (neuroblastoma RAS viral oncogene homolog) have been widely documented in various stages of human melanoma (8, 9). The BRAF V600E mutation, a substitution of valine to glutamic acid at codon 600, is one of the most frequent BRAF mutations in melanoma that drives constitutive MEK (mitogen-activated protein kinase kinases) activation, thereby leading to aberrant stimulation of extracellular signal-regulated kinase (ERK) activity (10).

BRAF mutations are known to be dominant drivers of melanoma, and hence targeting these mutations has been one of the success stories in treatment of cancer (11). However, resistance to BRAF inhibitors is rapidly acquired and is a major impediment to cancer therapy (12). Indeed, effectively blocking the key determinants of BRAF signaling could allow for a more sustained response. It is well known that mutations in BRAF per se lead to nevi that remain nonmalignant for many years or decades (13). What,

then, are the key rate-limiting determinants of BRAF signaling for progression to melanomas?

Mutations in *telomerase reverse transcriptase* (*TERT*) promoter have recently been described as predominant genetic alterations in over 70% of human melanomas (14, 15). Although the oncogenic role of BRAF V600E mutations as a major driver of melanoma progression have been well studied since its discovery in 2002, the significant association of BRAF and *TERT* promoter mutations was only recently reported in a high percentage of metastatic melanoma cases following the discovery of *TERT* promoter mutations in 2013 (16, 17). This co-occurrence of BRAF and *TERT* promoter alterations was also found in other cancer types such as papillary thyroid cancers and was highly correlated with worse prognosis of these cancers (17–19). *TERT* mutations occur in the core promoter region of the *telomerase reverse transcriptase* gene encoding the catalytic subunit of human telomerase and are known to generate a new motif for the E-twenty-six (ETS) family of transcription factors. Although several groups have characterized the major ETS factor(s) that bind mutant *TERT* promoters (20–23), it is not known if BRAF V600E-induced ERK signaling plays a key role in the transcriptional regulation of *TERT* reactivation at mutant *TERT* promoters in human melanoma.

In this study, we sought to investigate the mechanistic relevance of RAS-ERK activation in BRAF/NRAS mutant cells with mutant *TERT* promoters. Our findings demonstrate a major role of RAS-ERK signaling in the maintenance of an active

Significance

Mutations in BRAF are dominant drivers of melanoma, and hence targeting them has been a success story in treatment of cancer. However, resistance to BRAF inhibitors is rapidly acquired and is a major impediment to cancer therapy. BRAF mutations per se only lead to nevi that remain nonmalignant for decades. What then is the key rate-limiting determinant of BRAF signaling for progression to melanomas? We show that mutant *TERT* promoter is a key downstream target of RAS-ERK and its activation downstream of BRAF signaling is a major rate-limiting step in melanoma progression, suggesting that simultaneous disruption of BRAF signaling and active chromatin remodeling at mutant *TERT* promoters may be an attractive strategy to overcome BRAF-inhibitor resistance during melanoma therapy.

Author contributions: Y.L. and V.T. designed research; Y.L. and H.S.C. performed research; W.J.C. contributed new reagents/analytic tools; Y.L. and H.S.C. analyzed data; V.T. supervised the project; and Y.L. wrote the paper.

The authors declare no conflict of interest.

This article is a PNAS Direct Submission.

¹To whom correspondence may be addressed. Email: vinayt@imcb.a-star.edu.sg or liyh@imcb.a-star.edu.sg.

This article contains supporting information online at www.pnas.org/lookup/suppl/doi:10.1073/pnas.1611106113/-DCSupplemental.

chromatin state at mutant *TERT* promoters, which facilitates the recruitment of RNA polymerase II that activates *TERT* transcription in BRAF-mutant melanoma cells. ERK2's (mitogen-activated protein kinase 1) association at mutant *TERT* promoters promoted the dissociation of the histone deacetylase 1 (HDAC1) repressor complex from Sp1 sites, thereby leading to increased acetylation of histone 3 lysine 9 (H3K9ac). Hence, we provide evidence of the molecular reason for the significant coassociation of BRAF mutations and *TERT* promoter mutations in melanoma progression.

Results

BRAF/NRAS Mutant Melanoma Cell Lines Frequently Display *TERT* Promoter Mutations. To investigate a functional coassociation, if any, between BRAF/NRAS and *TERT* promoter mutations, we sequenced a variety of human melanoma cell lines for BRAF, NRAS, and *TERT* promoter mutations. From our analysis, we found seven melanoma cell lines harboring either BRAF V600E or NRAS mutations: A375, UACC257, G361, BLM, WM793, Malme-3M, and C8161 (Fig. 1*A* and *B*). Six BRAF/NRAS-mutant cell lines were found to contain *TERT* promoter mutations at positions -146C > T, -124C > T, or -138_139CC > TT (Fig. 1*A* and *B*), which result in the de novo creation of an ETS binding motif and are associated with increased transcriptional activity at the *TERT* promoter (14, 15). These observations are consistent with current literature reporting the frequent co-occurrence of BRAF/NRAS and *TERT* promoter mutations in human melanoma tumors (16, 17, 24, 25).

BRAF and MEK Inhibitors Reduce *TERT* Expression in BRAF/NRAS-Mutant Melanoma Cells. The C8161 cell line was recently reported to contain WT *TERT* promoter, thereby rendering it unresponsive to *TERT* inhibition when treated with MEK inhibitor (26). We thus analyzed the effect of RAS-ERK inhibition on *TERT* expression in other BRAF/NRAS-mutant cell lines carrying *TERT* promoter mutations. Short-term treatment of BRAF-mutant melanoma cell lines with the BRAF inhibitors vemurafenib and dabrafenib alone or in combination with the MEK inhibitor trametinib, as well as treatment of NRAS-mutant BLM cells with trametinib, resulted in the corresponding downregulation of MEK1/2 and ERK1/2 phosphorylation (Fig. 1*C*). Downstream reduction of ERK phosphorylation led to a rapid decline in *TERT* expression in all *TERT* promoter mutant melanoma cell lines following 1 d of treatment of BRAF and/or MEK inhibitors (Fig. 1*D*). This reduction in *TERT* expression corresponded with their decreased proliferation (Fig. S14), suggesting that RAS-ERK-mediated regulation of *TERT* transcription is important for the survival of melanoma cells.

Downregulation of RAS-ERK Signaling Inhibits the Transcriptional Activation of *TERT* at Mutant *TERT* Promoters in BRAF/NRAS-Mutant Melanoma Cells. Consistent with our findings using BRAF and MEK inhibitors, downregulation of RAS-ERK signaling via siRNA-mediated knockdown of BRAF in BRAF-mutant, *TERT* promoter mutant melanoma cells resulted in a similar decrease in *TERT* expression (Fig. 2*A*). These observations were further corroborated by luciferase reporter assays using reporter constructs for *TERT* promoters carrying either of the two hotspot mutations, -146C > T and -124C > T (14, 15), in three melanoma cell lines (Fig. 2*B*). Downregulation of ERK1/2 phosphorylation in BRAF-mutant A375 and UACC257 cells using siRNA targeting BRAF or in NRAS-mutant BLM cells via siRNA knockdown of ERK2 resulted in reduced transcriptional activation of both mutant *TERT* promoter reporters (Fig. 2*B*), demonstrating that the RAS-ERK pathway regulates transcriptional activation of *TERT* at *TERT* promoter mutations.

Reduction of ERK Phosphorylation Results in a Dramatic Loss of Telomerase Activity in BRAF/NRAS-Mutant Melanoma Cells. The reduction of phosphorylated ERK1/2 in BRAF/NRAS-mutant

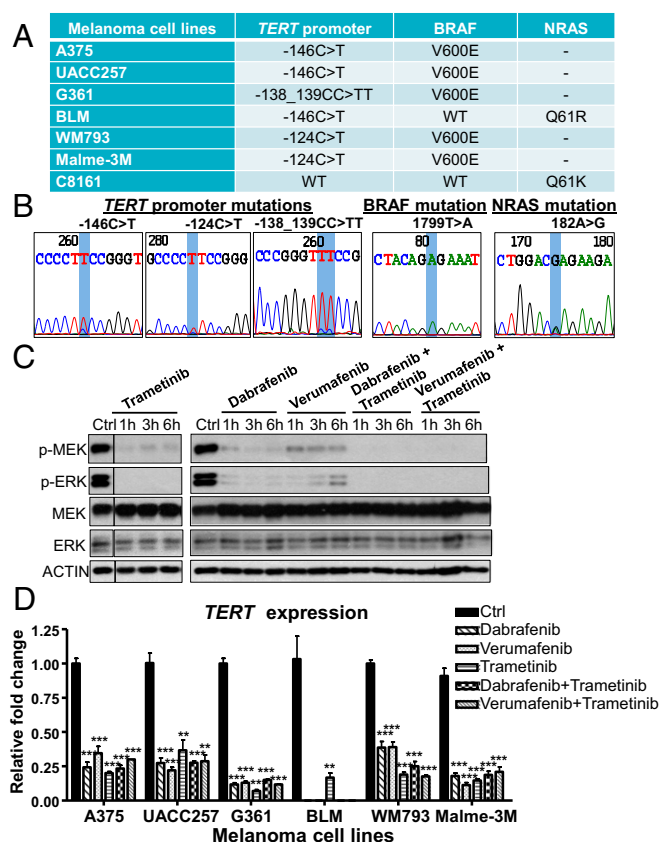


Fig. 1. Disruption of the RAS-ERK pathway inhibits the transcriptional activation of *TERT* at mutant *TERT* promoters in BRAF/NRAS mutant melanoma cells. (A) Table shows the profiles of BRAF, NRAS, and *TERT* promoter mutations in human melanoma cell lines. (B) Representative DNA chromatogram sequences of BRAF, NRAS, and *TERT* promoter mutations of melanoma cell lines analyzed. (C) A375 cells were treated with 0.4 μ M vemurafenib, 0.1 μ M dabrafenib, and 20 nM trametinib alone or in combination with trametinib for 1, 3, and 6 h. Cell lysates of control and treated samples were analyzed for changes in MEK1/2 and ERK1/2 phosphorylation by Western blotting. (D) BRAF/NRAS-mutant, *TERT* promoter mutant melanoma cells were treated with 0.1 μ M dabrafenib, 0.4 μ M vemurafenib, and 20 nM trametinib alone or in combination with trametinib for 1 d and analyzed for *TERT* expression by quantitative real-time PCR (qPCR). Data from three experiments are represented as the relative fold change in mRNA expression (mean $\pm$ SEM). Student's *t* test (two-tailed) was used for all statistical analyses: ***P* < 0.01; ****P* < 0.001.

melanoma cells carrying *TERT* promoter mutations, using either BRAF and MEK inhibitors (Fig. 1*C*) or si-BRAF treatment (Fig. S1*B*), both resulted in a dramatic loss of telomerase activity (Fig. 2*C* and *D*). This loss of telomerase activity correlated with the decreased long-term proliferation of shBRAF or shERK2 melanoma cells (Fig. 2*E* and *F*), which also display reduced expression of *TERT* and *urokinase plasminogen activator* (*uPA*), a RAS/ERK target gene (Fig. S24), following downregulation of BRAF (Fig. S2*B*). Re-expression of *TERT* in shBRAF or shERK2 melanoma cells (Fig. S2*C*) rescued the effect of RAS-ERK inhibition on cell proliferation (Fig. 2*E* and *F*), indicating that RAS-ERK-mediated regulation of *TERT* transcription is essential for the survival of melanoma cells. These findings therefore demonstrate the significant involvement of RAS-ERK signaling in the maintenance of telomerase expression and function in *TERT* promoter mutant melanoma cells.

RAS-ERK Signaling Regulates the Active Chromatin State of Mutant *TERT* Promoters. Because a dramatic epigenetic switch has been reported to underscore the activation of *TERT* at mutant *TERT*

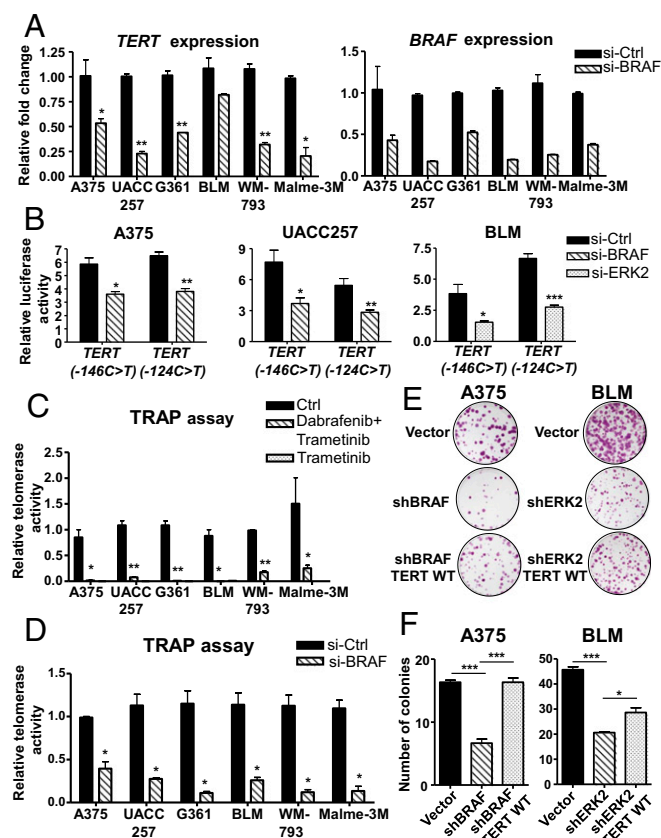


Fig. 2. Inhibition of RAS-ERK signaling reduces the telomerase activity and proliferation of BRAF/NRAS-mutant melanoma cells carrying *TERT* promoter mutations. (A) A375, UACC257, G361, BLM, WM793, and Malme-3M cells were treated with si-Control (Ctrl) or si-BRAF and analyzed for *TERT* and *BRAF* expression after 3 d. Plots represent relative fold changes in mRNA expression (mean $\pm$ SEM) from two independent experiments. (B) Luciferase reporter assays were performed in A375, UACC257, and BLM cells that were transfected with pGL3 reporter vector containing the *TERT* promoter region (-340 to -55) with -146C > T or -124C > T mutation. Transfected cells were treated with siRNA-targeting BRAF or ERK2. Data shown are representative of two independent experiments. (C) A375, UACC257, G361, WM793, and Malme-3M cells were treated with BRAF and MEK inhibitors (0.1 μ M dabrafenib and 20 nM trametinib), and BLM cells were treated with MEK inhibitor (20 nM trametinib). DMSO control (Ctrl) or BRAF/MEK inhibitor-treated cells were analyzed for telomerase activity after 2 d by a Telomere Repeat Amplification Protocol (TRAP) assay. Data shown are from two experiments and represent relative telomerase activity (mean $\pm$ SEM). (D) A375, UACC257, G361, BLM, WM793, and Malme-3M cells were transfected with si-Ctrl or si-BRAF and analyzed for telomerase activity after 3 d. Data from two experiments are shown (mean $\pm$ SEM). (E) Colony formation assay was performed in vector or shBRAF-expressing A375 and vector or shERK2 BLM cells. TERT was repressed in shBRAF and shERK2 cells via lentiviral transduction of Flag-TERT WT construct, and colony formation assay was performed in parallel. Representative images from two independent experiments are shown. (F) Plot depicts average number of colonies counted per field (mean $\pm$ SEM) from all experiments in the earlier assay. Student's *t* test (two-tailed) was used for all statistical analyses: **P* < 0.05; ****P* < 0.001.

promoters (22), we sought to investigate the molecular basis of the RAS-ERK pathway in the regulation of telomerase reactivation at mutant *TERT* promoters by performing chromatin immunoprecipitation (ChIP) assays using antibodies targeting various histone marks that are associated with active chromatin. Consistent with our earlier observations, downregulation of RAS-ERK signaling using BRAF and MEK inhibitors resulted in the reduction of trimethylation of histone H3 at lysine 4 (H3K4me3), a histone mark that is associated with transcriptional activation, at the mutant

TERT promoters of four melanoma cell lines (Fig. 3A). Moreover, various acetylated histone marks associated with active enhancers and transcriptional elongation, of which the H3K9ac is most prominently enriched, were substantially depleted at mutant *TERT* promoters following ERK inhibition (Fig. 3A). The reduction of H3K4me3 and H3K9ac levels at mutant *TERT* promoters was similarly observed in BRAF-mutant, *TERT* (-146C > T) or *TERT* (-124C > T) melanoma cell lines, UACC257, A375, WM793, and Malme-3M, following si-BRAF treatment (Fig. 3B). These data collectively demonstrate the critical role of RAS-ERK signaling in regulating the active chromatin state of mutant *TERT* promoters in BRAF/NRAS-mutant human melanoma.

Inhibition of the RAS-ERK Pathway Reduces the Recruitment of RNA Polymerase II at Mutant *TERT* Promoters in Human Melanoma Cells.

Consistent with the downregulation of an epigenetic signature for active chromatin, disruption of the RAS-ERK pathway by BRAF and MEK inhibitors led to the substantial reduction of RNA polymerase II (Pol II) recruitment at the mutant *TERT* promoters of BRAF/NRAS-mutant melanoma cell lines (Fig. 4A). Notably, specific enrichment of serine 5 phosphorylated Pol II (Pol II S5P), which is most highly enriched and associated with transcriptional activation at gene promoters, was substantially depleted at the mutant *TERT* promoters of BRAF/MEK inhibitor-treated

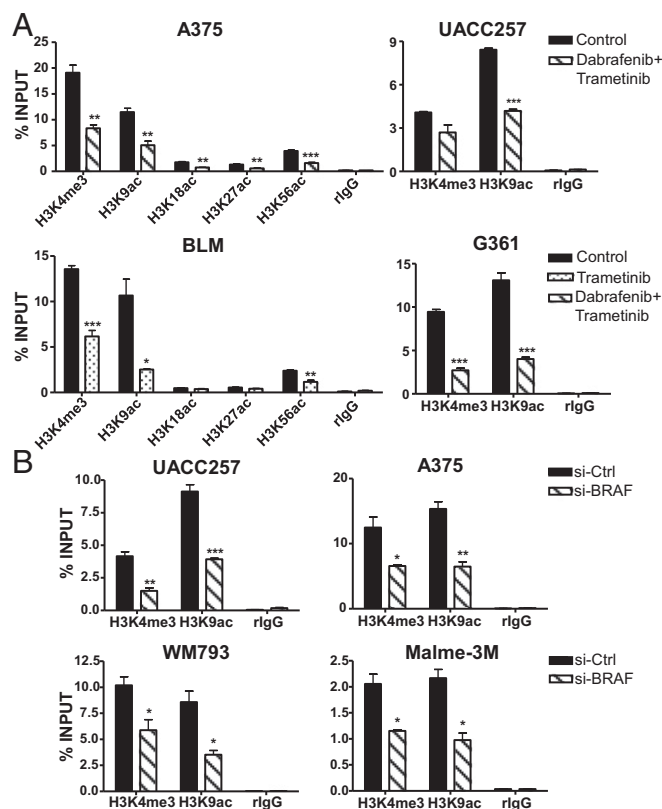
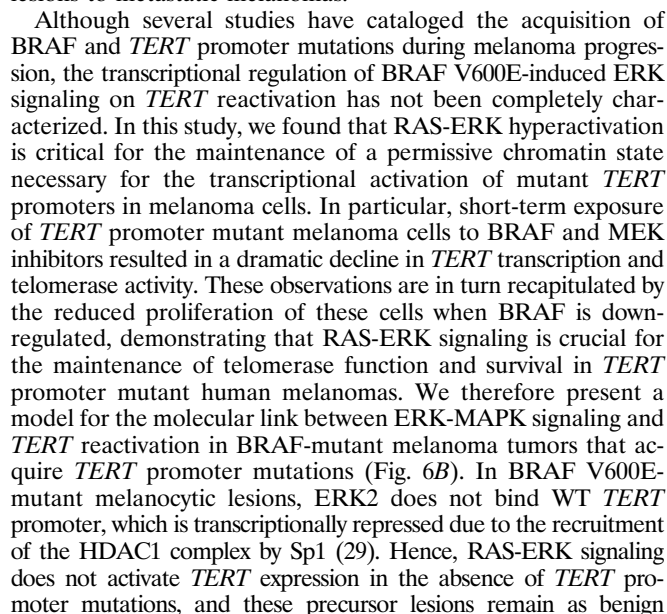


Fig. 3. RAS-ERK signaling regulates the active chromatin state of mutant *TERT* promoters. (A) ChIP was performed in control or BRAF/MEK inhibitor-treated A375, UACC257, BLM, and G361 cells using antibodies specific for different histone marks and IgG as negative control. Enrichment of *TERT* promoter DNA fragments in ChIP DNA was normalized to DNA input. (B) ChIP was performed in si-Ctrl or si-BRAF-transfected UACC257, A375, WM793, and Malme-3M cells using antibodies specific for H3K4me3 and H3K9ac and IgG as negative control. Data shown represent at least two independent ChIP assays for all cell lines. Student's *t* test (two-tailed) was used for all statistical analyses of control versus treated samples for each ChIP: **P* < 0.05; ***P* < 0.01; ****P* < 0.001.



melanocytic nevi (Fig. 6B). However, when BRAF mutant tumors attain *TERT* promoter mutations, RAS-ERK hyperactivation leads to the association of ERK2 to mutant *TERT* promoter, potentially through interaction with GABPA or ETS1 (34, 35), and its subsequent phosphorylation of Sp1 stabilizes it on the mutant promoter. Stabilized ERK on mutant *TERT* promoter facilitates the dissociation of HDAC1 from mutant *TERT* promoters, thereby promoting histone 3 lysine 9 acetylation, which generates an active chromatin state that mediates *TERT* reactivation (Fig. 6B). This results in the sustained telomerase activity of tumor cells that promotes their development to malignant melanomas through canonical and noncanonical functions of *TERT* (Fig. 6B) (23, 36–39).

One of the major challenges faced in the effective therapy and management of metastatic melanomas is their acquired resistance to BRAF and MEK inhibitors (40). Our findings reveal that the mutant *TERT* promoter is a major substrate for the RAS-ERK pathway and is a key determinant for the progression of BRAF mutant lesions to malignant melanomas. Hence, combination therapy involving BRAF/MEK inhibitors and blockers of chromatin remodeling of mutant *TERT* promoters may be an attractive strategy to inhibit telomerase reactivation in melanoma

cancers with BRAF/NRAS and *TERT* promoter mutations. It will also be important to further define the transcription factors and epigenetic regulators that maintain the active chromatin state of mutant *TERT* promoters downstream of ERK signaling in cancers.

Materials and Methods

Human melanoma G361 and BLM cell lines were kindly provided by Birgit Lane (Institute of Medical Biology, Singapore). A375, UACC257, and Malme-3M cell lines were obtained from Ernesto Guccione (Institute of Molecular and Cell Biology, Singapore). The WM793 cell line was a gift from Jean-Pierre Abastado (Singapore Immunology Network, Agency for Science Technology and Research, Singapore). C8161 melanoma cells were given by Nicolas Dumaz (INSERM, U976, Skin Research Center, Hôpital Saint-Louis, Paris) (26). Details of the reagents and experimental procedures of various assays used for these cell lines are provided in *SI Materials and Methods*. Institutional review board approval or informed consent was not necessary for this study as research did not involve humans, human embryonic stem cells, or vertebrate animals.

ACKNOWLEDGMENTS. We thank the Agency for Science Technology and Research (Singapore) and the Institute of Molecular and Cell Biology for their support and funding of this work.

- Paluncic J, et al. (2016) Roads to melanoma: Key pathways and emerging players in melanoma progression and oncogenic signaling. *Biochim Biophys Acta* 1863(4): 770–784.
- Johannessen CM, et al. (2013) A melanocyte lineage program confers resistance to MAP kinase pathway inhibition. *Nature* 504(7478):138–142.
- Poulidakos PI, Rosen N (2011) Mutant BRAF melanomas: Dependence and resistance. *Cancer Cell* 19(1):11–15.
- Wellbrock C, Arozarena I (2016) The complexity of the ERK/MAP kinase pathway and the treatment of melanoma skin cancer. *Front Cell Dev Biol* 4:33.
- Davies H, et al. (2002) Mutations of the BRAF gene in human cancer. *Nature* 417(6892):949–954.
- Garnett MJ, Marais R (2004) Guilty as charged: B-Raf is a human oncogene. *Cancer Cell* 6(4):313–319.
- Bos JL (1989) ras oncogenes in human cancer: A review. *Cancer Res* 49(17):4682–4689.
- Poynter JN, et al. (2006) BRAF and NRAS mutations in melanoma and melanocytic nevi. *Melanoma Res* 16(4):267–273.
- Shain AH, et al. (2015) The genetic evolution of melanoma from precursor lesions. *N Engl J Med* 373(20):1926–1936.
- Wan PT, et al.; Cancer Genome Project (2004) Mechanism of activation of the RAF-ERK signaling pathway by oncogenic mutations of B-Raf. *Cell* 116(6):855–867.
- Flaherty KT, et al. (2012) Combined BRAF and MEK inhibition in melanoma with BRAF V600 mutations. *N Engl J Med* 367(18):1694–1703.
- Merlino G, et al. (2016) The state of melanoma: Challenges and opportunities. *Pigment Cell Melanoma Res* 29(4):404–416.
- Michaloglou C, et al. (2005) BRAF^{V600E}-associated senescence-like cell cycle arrest of human naevi. *Nature* 436(7051):720–724.
- Horn S, et al. (2013) TERT promoter mutations in familial and sporadic melanoma. *Science* 339(6122):959–961.
- Huang FW, et al. (2013) Highly recurrent TERT promoter mutations in human melanoma. *Science* 339(6122):957–959.
- Heidenreich B, et al. (2014) Telomerase reverse transcriptase promoter mutations in primary cutaneous melanoma. *Nat Commun* 5:3401.
- Nagore E, et al. (2016) TERT promoter mutations in melanoma survival. *Int J Cancer* 139(1):75–84.
- Jin L, et al. (2016) BRAF and TERT promoter mutations in the aggressiveness of papillary thyroid carcinoma: A study of 653 patients. *Oncotarget* 7(14):18346–18355.
- Liu X, et al. (2014) TERT promoter mutations and their association with BRAF V600E mutation and aggressive clinicopathological characteristics of thyroid cancer. *J Clin Endocrinol Metab* 99(6):E1130–E1136.
- Bell RJA, et al. (2015) Cancer. The transcription factor GABP selectively binds and activates the mutant TERT promoter in cancer. *Science* 348(6238):1036–1039.
- Liu Y, et al. (2015) Non-canonical NF- κ B signalling and ETS1/2 cooperatively drive C250T mutant TERT promoter activation. *Nat Cell Biol* 17(10):1327–1338.
- Stern JL, Theodorou D, Vogelstein B, Papadopoulos N, Cech TR (2015) Mutation of the TERT promoter, switch to active chromatin, and monoallelic TERT expression in multiple cancers. *Genes Dev* 29(21):2219–2224.
- Li Y, Tergaonkar V (2016) Telomerase reactivation in cancers: Mechanisms that govern transcriptional activation of the wild-type vs. mutant TERT promoters. *Transcription* 7(2):44–49.
- Griewank KG, et al. (2014) TERT promoter mutation status as an independent prognostic factor in cutaneous melanoma. *J Natl Cancer Inst* 106(9):dju246.
- Macerola E, et al. (2015) Coexistence of TERT promoter and BRAF mutations in cutaneous melanoma is associated with more clinicopathological features of aggressiveness. *Virchows Archiv* 467(2):177–184.
- Vallarelli AF, et al. (2016) TERT promoter mutations in melanoma render TERT expression dependent on MAPK pathway activation. *Oncotarget* 7(33):53127–53136.
- Akincilar S, et al. (2016) Long-range chromatin interactions drive mutant TERT promoter activation. *Cancer Discov* CD-16-0177.
- Takakura M, et al. (2001) Telomerase activation by histone deacetylase inhibitor in normal cells. *Nucleic Acids Res* 29(14):3006–3011.
- Won J, Yim J, Kim TK (2002) Sp1 and Sp3 recruit histone deacetylase to repress transcription of human telomerase reverse transcriptase (hTERT) promoter in normal human somatic cells. *J Biol Chem* 277(41):38230–38238.
- Liao M, Zhang Y, Dufau ML (2008) Protein kinase C α -induced derepression of the human luteinizing hormone receptor gene transcription through ERK-mediated release of HDAC1/Sin3A repressor complex from Sp1 sites. *Mol Endocrinol* 22(6): 1449–1463.
- Göke J, Chan Y-S, Yan J, Vingron M, Ng H-H (2013) Genome-wide kinase-chromatin interactions reveal the regulatory network of ERK signaling in human embryonic stem cells. *Mol Cell* 50(6):844–855.
- Tee W-W, Shen SS, Oksuz O, Narendra V, Reinberg D (2014) Erk1/2 activity promotes chromatin features and RNAPII phosphorylation at developmental promoters in mouse ESCs. *Cell* 156(4):678–690.
- Chen Y-J, Wang Y-N, Chang W-C (2007) ERK2-mediated C-terminal serine phosphorylation of p300 is vital to the regulation of epidermal growth factor-induced keratin 16 gene expression. *J Biol Chem* 282(37):27215–27228.
- Flory E, Hoffmeyer A, Smola U, Rapp UR, Bruder JT (1996) Raf-1 kinase targets GAB-binding protein in transcriptional regulation of the human immunodeficiency virus type 1 promoter. *J Virol* 70(4):2260–2268.
- Selvaraj N, Kedage V, Hollenhorst PC (2015) Comparison of MAPK specificity across the ETS transcription factor family identifies a high-affinity ERK interaction required for ERG function in prostate cells. *Cell Commun Signal* 13:12.
- Akincilar SC, Unal B, Tergaonkar V (2016) Reactivation of telomerase in cancer. *Cell Mol Life Sci* 73(8):1659–1670.
- Khattar E, et al. (2016) Telomerase reverse transcriptase promotes cancer cell proliferation by augmenting tRNA expression. *J Clin Invest* 126(10):4045–4060.
- Li Y, Tergaonkar V (2014) Noncanonical functions of telomerase: Implications in telomerase-targeted cancer therapies. *Cancer Res* 74(6):1639–1644.
- Koh CM, et al. (2015) Telomerase regulates MYC-driven oncogenesis independent of its reverse transcriptase activity. *J Clin Invest* 125(5):2109–2122.
- Samatar AA, Poulidakos PI (2014) Targeting RAS-ERK signalling in cancer: Promises and challenges. *Nat Rev Drug Discov* 13(12):928–942.
- Akincilar SC, et al. (2015) Quantitative assessment of telomerase components in cancer cell lines. *FEBS Lett* 589(9):974–984.